

Your Article Title

Your Name

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Abstract

Your abstract goes here. This is a brief summary of your article.

1 Introduction

This paper presents the methodology to classify neighbourhoods based on the observed mobility behaviours of their residents. Similar to existing geodemographic classifications (?).

In this paper we employ Multinomial Regression and Poststratification (MRP) to map transport behaviours at small area level.

2 Background

2.1 Building area-level profiles from individual survey responses

It is not uncommon that, when using survey data, we find ourselves in the situation where the data has been captured at a specific spatial level, but we want to draw estimations at a higher spatial resolution level. In these cases, the survey will not be representative, since it has been designed for a coarser spatial resolution. This results in areas not having enough observations for deriving robust estimations. This is a typical problem of small-area estimation. Multilevel regression and poststratification (MRP) is an increasingly popular approach to deriving robust estimates from non-representative survey samples at small area level. MRP has traditionally been used in polling to estimate voting counts by constituency.

Multilevel regression and poststratification (MRP) is an increasingly popular tool for adjusting a non-representative sample to a larger population. In particular, MRP appears to be effective in areas where conventional design-based survey approaches have traditionally struggled, notably small-area estimation (Rao, 2014; Pfeffermann et al., 2013; Zhang et al., 2014) and with convenience sampling (Wang et al., 2015).

One difference between MRP and traditional poststratified design-based weights is that MRP uses partial pooling. Simple poststratification has difficulties with empty cells, in which case the usual practice is to poststratify only on marginals (thus ignoring interactions), or pool cells together. In contrast, the partial pooling of multilevel modeling automatically regularizes group estimates.

3 Methods

We selected a set of demographic variables (Table 1) to describe the personal and household-level characteristics of the individuals in the sample.

NTS Variable	Type	Description
Individual-level demographic variables		
Sex_B01ID	Binary	Sex of the respondent (male / female).
Age_B01ID	Ordinal	Age of the respondent in 21 banded categories, ranging from under 1 year to 80 and over.
MaritalStat_B01ID	Categorical	Legal marital status, including never married, married or in a civil partnership, separated, divorced, and widowed.
EthGroupTS_B02ID	Binary	Ethnic group, time-series consistent classification: White; Non-White.
EcoStat_B02ID	Categorical	Economic activity status in 6 categories: employed full-time; employed part-time; unemployed; economically inactive (retired or permanently sick); economically inactive (student); economically inactive (other).
EdAttn3_B01ID	Ordinal	Highest educational qualification attained, ranging from no qualifications to degree level and above.
Caretime_B01ID	Ordinal	Hours per week spent providing unpaid care for a family member or friend, in banded categories.
Household-level demographic variables		
HHoldGOR_B02ID	Categorical	Government Office Region of the household's usual residence.
HHoldStruct_B02ID	Categorical	Household structure in 6 categories, including one-person households, couples with and without dependent children, and lone-parent households.
HHoldNumChildren	Count	Number of children (aged under 16) in the household.
HHoldNumPeople	Count	Total number of people living in the household.
AddressType_B01ID	Categorical	Type of dwelling (detached, semi-detached, terraced, flat or maisonette, and other).
Ten1_B02ID	Categorical	Tenure type in 3 categories: owned or shared ownership; social rented; private rented.
HLongAN_B01ID	Ordinal	Length of time the household has lived at the current address, in banded categories.
NumCar	Count	Number of cars available to the household (excludes vans and other light goods vehicles).
HHIncQDS2008_B01ID	Ordinal	Equivalised household income quintile derived from the diary sample, using 2008 income bandings for time-series consistency.

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Table 1 continued from previous page

NTS Variable	Type	Description
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Table 1: Variables used in the analysis, sourced from the National Travel Survey (NTS) 2024. Demographic variables are drawn from the individual and household interview files. Travel behaviour variables are derived from the 7-day travel diary unless otherwise indicated.

3.0.1 Multilevel Regression with Post-stratification

One of the most well-known uses of MRP is the generation of electoral forecasts using non-representative polls. Intuitively, MRP works in two stages. First, we need to gather individual-level information from a survey with information on both our variable of interest and demographic and socioeconomic attributes of the respondents. Using this data, we estimate a model that predicts the variable of interest based on the socio-demographic attributes at individual level. This step helps identify which demographic subgroups are more likely to present the variable of interest. In the second step, we gather aggregated data about the small areas of interest, usually from the census, and estimate the presence of those identified subgroups at the small area level. This allows us to estimate the prevalence of the variable of interest in the area weighted by the frequency of the subgroups at the unit of interest.

4 Results

Present your results.

4.1 A Subsection

You can add subsections as needed.

5 Discussion

Discuss your findings.

6 Conclusion

Your conclusions.